Unit 3 - Logic

Section 3.1: Introduction to Reasoning

Before we can study logic, we need to understand what it is we're studying. What does it mean to reason? What is an argument? What is an argument made of, and how can we tell that we are seeing one?

In this section, we will define our initial terms and learn to recognize various forms of reasoning. _____

Concept 1 – What Is an Argument?

When we talk about logic,we will be concerned with *arguments*. We need to take a moment to understand what that is.

Definition: Argument
An **argument** is an organized sequence of statements designed to lay out a line of reasoning.

That definition doesn't get us very far, though. To understand it, we need a good definition for *reasoning*.

Definition: Reasoning
Reasoning is an _____ thought process designed to find _____ based on_____ .

We use reasoning a *lot* in math class. In particular, we use a kind of reasoning called **mathematical reasoning**. In mathematical reasoning, we use a set of rules—the rules of algebra—to learn new facts about numbers using known facts about those numbers.

Example 1: We use mathematical reasoning to solve the equation $2x + 3 = 7$ and find that $x = 2$. To make an *argument* for this solution, we need to lay out our reasoning.

Statement	Justification
$2x + 3 = 7$	
$2x = 4$	
$x = 2$	

This is an argument, because it lays out the _____ of our thinking.

For the rest of this unit, we will be looking at *verbal* arguments, rather than mathematical ones, but keep this example in mind to see that this isn't so different from what we normally do in math class.

Concept 2 – Statements

At its core, an argument is a series of *statements.*

Definition: Statement
A **statement** is a sentence that can be either true or false. A statement is **true** if it corresponds to reality, it's **false** if it doesn't correspond to reality.

Not every sentence is a statement.

Example 2: The following are types of sentences that are not statements.

- "What time is it?" _____
- "Turn to page 378." _____
- "Oh, no!" _____

Example 3: There are also some sentences that *look* like statements, but we can't make sense of them.

- "The number 7 melts at 35° Celsius." _____

  _____

- "This sentence is false." _____

  _____

We can also have sentences that are poorly defined: we can't tell if they're _____ or _____ because it's hard to say what they mean.

Example 4: Consider the sentence, "Yellow Submarine is the best Beatles album."

_____

This sentence involves an _____. We can make it a statement by restating it in more concrete terms.

- Yellow Submarine is Alice's favorite Beatles album.

  This is now a statement about _____.

- Yellow Submarine sold more copies than any other Beatles album.

  We can tell whether or not this is true if we look up _____ .

Note: When we study logic, we sometimes allow ambiguous statements (for example, "John is tall,") in order to speak simply. When we do that, we assume the terms are clearly defined *somewhere*.

Concept 3 – Premises and Conclusions

An argument consists of two kinds of statements:

- The known facts we start with are called _____ .
- The new facts we arrive at are called _____ .

In a complicated argument, there might be many premises and several smaller conclusions, leading up to the main conclusion. The kind of arguments we will analyze in this class will usually be fairly simple, however, with two or three premises and only one conclusion.

Example 5: In this example argument, identify the premises and conclusion.

Fozzie is a bear. As a rule, bears are not funny. Therefore, Fozzie is not funny.

First, we'll identify the conclusion. It's the fact this argument is trying to convince us is true.

- _____

Next, we'll identify the premises. These are the facts that are supposed to led us to this conclusion.

- _____
- _____

In Example 5, the word _____ indicates the conclusion. There are a variety of words we can use in English when we want to point to the conclusion of an argument.

_____

_____

_____

Also in Example 5, the phrase _____ indicates that one of the statements is a premise. We can use a variety of phrases in English to indicate premises, too.

_____

_____

_____

Concept 4 – Types of Reasoning

We've already mentioned one type of reasoning, *mathematical reasoning*.

Definition: Mathematical Reasoning
In **mathematical reasoning**, the premises are statements about numbers, and the rules of _____ are used to make conclusions that are also about numbers.

Mathematical reasoning is a specialized type of *deductive reasoning*.

Definition: Deductive Reasoning
In **deductive reasoning**, the premises include one or more general rules, and the conclusion either combines or applies these rules. In deductive reasoning, we reason from the _____ to the _____ .

In mathematical reasoning, we include a set of general rules, the rules of algebra, as extra premises.

Example 6: Explain why this argument is an example of deductive reasoning.

If it rains on Saturday, I do the laundry. This Saturday, it rained. Therefore, I did the laundry.

In order to analyze an argument, we must first identify the _____ and the _____.

The conclusion is:

Conclusion: _____

The remaining statements are the premises. They are:

Premise 1: _____

Premise 2: _____

One of these, Premise _____ is a general statement about what I do on Saturdays. The conclusion _____ this rule to a _____ Saturday.

Deductive reasoning is often contrasted with *inductive reasoning*.

Definition: Inductive Reasoning

In **inductive reasoning**, the premises are all specific observations, and the conclusion is either a general rule derived from those statements, or an application of such a rule to cases not observed.

In inductive reasoning, we reason from the _____ to the _____ .

Example 7: Explain why this argument is an example of inductive reasoning.

> I have eaten at this restaurant three times. Every time, I got sick. Therefore, I will get sick any time I eat at this restaurant.

The conclusion of this argument is a general statement:

Conclusion: _____

The remaining statements are the premises. They are:

Premise 1: _____

Premise 2: _____

Both of these premises are statements about the _____ I've eaten at the restaurant.

Since the conclusion is a _____ , but the premises are _____, this argument is inductive.

Notice that the premises of this inductive argument aim to establish a _____ which the conclusion then _____.

Example 8: Explain why this argument is an example of inductive reasoning.

> For the last two days, traffic has been bad on my way to work. Therefore, I expect traffic will be bad on my way to work again today.

In this argument, the conclusion is:

Conclusion: _____

There is only one premise.

Premise 1: _____

The premise establishes the pattern that the traffic is _____ or _____ bad on my way to work.

The conclusion _____ this pattern to a day that wasn't (yet) observed. The argument doesn't explicitly *describe* the pattern, but since the conclusion applies it, the argument is inductive.

Concept 5 – Valid Arguments

We have a notion of an argument "working," which we call being *valid*. Because they operate a little differently, we have different definitions of validity for deductive and inductive arguments.

Definition: Valid (deductive)
A *deductive* argument is **valid** if knowing that the premises are _____ guarantees that the conclusion is _____. In other words, an argument is **invalid** (not valid) if its conclusion can be _____ while its premises are all _____.

We will learn a lot of techniques for determining whether a deductive argument is valid later in this unit! For now, though, we'll analyze an example of deductive reasoning just by looking at its meaning.

Example 9: The following is a valid deductive argument.

> *Premise 1:* All frogs are green.
> *Premise 2:* Kermit is a frog.
> *Conclusion:* Therefore, Kermit is green

To see that this is valid, we'll imagine what it would be like if the conclusion was false.

What if Kermit wasn't green? Say Kermit is blue. Then there are two possibilities.

- Maybe Kermit is _____ . So premise ____ is false.
- Otherwise, Kermit is _____, so _____ . So premise ____ is false.

Since making the _____ false forces at least one _____ to be false, this argument is valid.

Let's compare this to an invalid deductive argument.

Example 10: The following deductive argument is invalid.

> *Premise 1:* Some dogs are gray.
> *Premise 2:* Some dogs are big.
> *Conclusion:* Therefore, all dogs are either big or gray.

We've probably seen gray dogs, and big dogs, so we know that in real life, both premises are _____.

In order for the conclusion to be false, we need _____ _____ . But we've probably also seen, for example, a _____. So in real life, the conclusion is false.

When we want to decide whether a _____ argument is valid, we try to find a way the conclusion can be _____ while the premises are _____.

We have a slightly different definition of validity inductive arguments.

Definition: Valid (inductive)
An *inductive* argument is **valid** if knowing that the premises are _____ makes the conclusion _____ to be _____.

This gives us less _____ in inductive arguments than in deductive ones.

Example 11: Let's consider a valid inductive argument.

> *Premise 1:* I've never seen a purple cat.
> *Premise 2:* I've seen a lot of cats.
> *Conclusion:* Therefore, there's no such thing as a purple cat.

Notice that both premises are _____ (about cats that _____), and the conclusion is _____ (about all cats).

But the second premise says that, if there were any purple cats, I _____ would have seen one!

This shows the main difference between inductive and deductive arguments.

- In a deductive argument, if the _____ are true, the _____ is _____ to be true.
- But in an inductive argument, if the _____ are true, the _____ is only _____ true.

Why bother with inductive arguments, if they can't guarantee that their conclusions are true? In real life, we can only observe _____ facts; we can't *see* _____!

However, for most of the rest of this unit, we will focus on deductive arguments.

Concept 6 – Sound vs. Valid

Have a look again at the valid deductive argument we saw in Example 9.

> *Premise 1:* All frogs are green.
> *Premise 2:* Kermit is a frog.
> *Conclusion:* Therefore, Kermit is green

We saw that this argument is *valid*, that is if the _____ are all true, the _____ *must* be true. But there's something weird here…

Premise 1: Are all frogs *really* green?

_____

Premise 2: Is Kermit the Frog *really* a frog?

_____

Conclusion: But… is Kermit green?

_____

In this example, it turns out that both premises are _____, but the conclusion is still _____! This can happen with a valid argument. Remember the definition:

Definition: Valid
A *deductive* argument is **valid** if knowing that the premises are true *guarantees* that the conclusion is also true. An *inductive* argument is **valid** if knowing that the premises are true makes the conclusion *very likely* to be true.

If one or more of the premises are _____, what does this tell us about the conclusion? _____

We like this, because when we *analyze* an argument, we often like to take away the specific statements and just study its structure. However, we also want to be able to describe the kind of argument that *actually* has to have a true conclusion!

Definition: Sound
An argument is **sound** if it is _____ and its premises are all _____.

This gives us what we really want an argument to do:

- If a _____ argument is sound, then its conclusion is *guaranteed* to be true.
- If an _____ argument is sound, then its conclusion is *probably* true.